

# Enhanced Design Report

## Introduction

This project is an Interactive Generative AI Web App designed to engage users with the beauty of Chinese poetry. The app enables users to upload an image, select a type of Chinese poem, and receive a generated poem based on their inputs. Utilizing a Chinese poetry dataset from Hugging Face and LangChain's Retrieval-Augmented Generation (RAG) capabilities, the app enhances the learning experience by connecting visual and literary art forms through AI.

## Choice of Models

1. **Hugging Face Image-to-Text Model:** This model translates visual elements into descriptive text, providing a bridge between the uploaded image and the generated poem. It captures the essence of the visual input and conveys it as text, which serves as the basis for poetic creation.
2. **Hugging Face Model (Chinese Poem Dataset):** The dataset offers historical Chinese poetry, enabling the app to generate poetry that is stylistically and contextually in line with traditional Chinese forms.
3. **LangChain for Retrieval-Augmented Generation (RAG):** LangChain's RAG capabilities allow the app to draw relevant content from the poetry dataset, enriching each generated poem with accurate historical context.
4. **OpenAI Model (GPT-4):** GPT-4 adds a creative touch by structuring responses into coherent and stylistically appropriate poetry based on the descriptive text and poetic form selected by the user.

## Rationale

The decision to use RAG and GPT-4 in conjunction with the Hugging Face dataset aligns with the app's goal of promoting engagement with Chinese poetry. The integration of these models offers a unique pathway for users to learn about different poetic structures and experience the generative capabilities of AI within a culturally enriching context. By combining these models, the app also demonstrates how AI can facilitate a deeper, dynamic interaction with literary art.

## Challenges and Solutions

Integrating LangChain was challenging, particularly in fine-tuning retrieval to align with Chinese poetic forms. Finding a suitable Chinese poetry dataset also required effort, as it involved data cleaning to ensure compatibility and accuracy. Iterative adjustments in prompt templates helped streamline the integration, allowing all models to work cohesively within the app.

## Inspiration

Inspired by Allison Parrish's AI poetry experiments, this project leverages AI to interpret visual stimuli and generate poetry. This approach honors Parrish's work on machine learning in literary creativity, pushing further into cross-media artistic experiences by focusing on Chinese poetry's rich traditions.

## Future Integration

Future improvements will explore how to handle different reading styles for various forms of Chinese poetry, considering aspects such as rhythm, tone, and punctuation. By refining the app's understanding of how each poem type is read, future versions could provide users with an enhanced auditory experience that aligns with the poem's structure.